

One-Page Individual Reflection

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Role in Project: Dataset & Preprocessing Pipeline

Project: CNN-Based Traffic Sign Recognition System — MLA0408 (Deep Learning)

1. What design/development decisions did you personally make or contribute to, and why?

My main contribution to the project was developing the CNN-based traffic sign recognition system and preparing the traffic sign images for classification. I contributed to selecting the GTSRB dataset because it contains different categories of traffic signs suitable for training a deep learning model. I also worked on image preprocessing by resizing the images to a fixed size and normalizing the pixel values so that the CNN could receive consistent input. I contributed to selecting CNN layers such as convolution, ReLU activation, pooling, flatten and fully connected layers because these layers help the model learn important features such as edges, shapes, colours and patterns from traffic signs. I also supported the use of data augmentation to improve the model's ability to recognize traffic signs under different lighting, orientations and image conditions.

2. What challenges did you face (technical or teamwork), and how did you resolve them?

The main technical challenge I faced was handling traffic sign images with different sizes, orientations and visual conditions. If the images were directly provided to the CNN without proper preprocessing, it could affect the training and classification performance. I resolved this by resizing all images to a fixed dimension and normalizing the pixel values before training. Another challenge was ensuring that different traffic sign classes were properly represented during training, so I checked the dataset and used suitable training, validation and testing splits. From the teamwork side, I communicated with my team members about the dataset, preprocessing and model requirements so that the prepared data could be used correctly for CNN training and testing.

3. What did you learn from this assignment beyond what was directly taught in class?

Through this assignment, I learned that the performance of a deep learning model depends not only on the CNN architecture but also on the quality and preparation of the input data. I gained practical knowledge about image preprocessing, normalization, data augmentation and dataset splitting. I also understood how CNN layers gradually learn different visual features, starting from simple edges and patterns and moving towards more complex shapes of traffic signs. This assignment helped me understand how theoretical concepts such as convolution, pooling, activation functions and classification can be applied to solve a real-world computer vision problem.

4. How does this project connect to SDG 9 and/or SDG 11?

This project is mainly connected to ****SDG 9 – Industry, Innovation and Infrastructure****, because the CNN-based traffic sign recognition system uses artificial intelligence and deep learning to support intelligent transportation and driver-assistance technologies. A reliable traffic sign recognition system can become an important part of smart transportation infrastructure. The project is also connected to ****SDG 11 – Sustainable Cities and Communities****, because accurate recognition of traffic signs can support safer road transportation and smarter urban mobility. Such technology can help vehicles understand road signs and improve road safety in future intelligent transportation systems.

5. How did this assignment help you attain CO1 / CO2 / CO3?

This assignment helped me attain **CO1** by improving my understanding of fundamental deep learning concepts such as CNN architecture, convolution, activation functions, pooling, preprocessing and image classification. It helped me attain **CO2** by giving me practical experience in implementing a CNN-based image classification system using a deep learning framework such as TensorFlow/Keras and working with a real traffic sign dataset. It also supported **CO3** by requiring me to evaluate the model's performance and understand how factors such as dataset quality, preprocessing, augmentation and CNN architecture influence classification results. Overall, the assignment helped me connect the concepts learned in class with a practical real-world application.

6. If given more time/resources, what would you improve and why?

If I had more time and resources, I would improve the project by experimenting with advanced CNN architectures such as ResNet, EfficientNet and MobileNet and comparing their performance with the basic CNN model. I would also try advanced data augmentation techniques to improve recognition under challenging conditions such as poor lighting, blur, rotation and partially hidden traffic signs. Another improvement would be to convert the current image-based system into a real-time traffic sign recognition system using camera or video input. I would also consider integrating an object detection model so that the system could first locate a traffic sign in a road scene and then classify it, making the project more suitable for intelligent transportation and driver-assistance applications.